

## Quantum-Assisted Multi-Asset Portfolio Construction

# Step 10

### **Risk-Adjusted Challenge Scoring**

Selecting zero-hard-breach leaders with transparent risk-adjusted evidence

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# 1 The challenge rule as a mathematical decision process

The rule is: the best risk-adjusted outcome with zero hard-constraint breaches receives the highest score. Step 10 translates that sentence into two sequential operations:

1. an eligibility gate that cannot be offset by return; and
2. a risk-adjusted ranking among eligible portfolios.

## 2 Eligibility gate

For candidate  $i$ , define

$$\mathcal{E}_i = \mathbf{1}\{H_i = 0\}\mathbf{1}\{U_i = 1\}\mathbf{1}\{C_i = 1\}, \quad (1)$$

where  $H_i$  is hard-breach count,  $U_i$  indicates a unique weight vector, and  $C_i$  indicates a challenge candidate. The challenge-wide screen includes every unique zero-breach candidate, including the two investable Step 4 portfolios. A second, fully certified set additionally requires independent Step 7 validation and common forward evidence.

A hard-breaching portfolio is disqualified; it is not assigned a lower numerical score that could be offset by high expected return.

## 3 Net expected return

For a one-year horizon,

$$R_i^{\text{net}} = R_i - C_i, \quad (2)$$

where  $C_i$  is the initial modeled implementation cost expressed in return-percentage units. For horizon  $H$  years, a simple sensitivity is

$$R_{i,H}^{\text{net}} = R_i - \frac{C_i}{H}. \quad (3)$$

This assumes the cost is paid once and is not a recurring annual rebalance cost.

## 4 Core risk-adjusted ratios

With no declared benchmark rate, the provisional screen uses

$$A_i^\sigma = \frac{R_i^{\text{net}}}{\sigma_i}, \quad (4)$$

$$A_i^S = \frac{R_i^{\text{net}}}{L_i^{\text{worst}}}, \quad (5)$$

$$A_i^D = \frac{R_i^{\text{net}}}{\text{MDD}_i}. \quad (6)$$

These are not called Sharpe or Calmar ratios because no common risk-free rate is fixed and the drawdown is model-based rather than a historical CAGR drawdown measure.

## 5 Normalization and score

Each positive ratio is normalized to the best candidate value:

$$\tilde{A}_i^k = \frac{A_i^k}{\max_{j \in \mathcal{E}} A_j^k}. \quad (7)$$

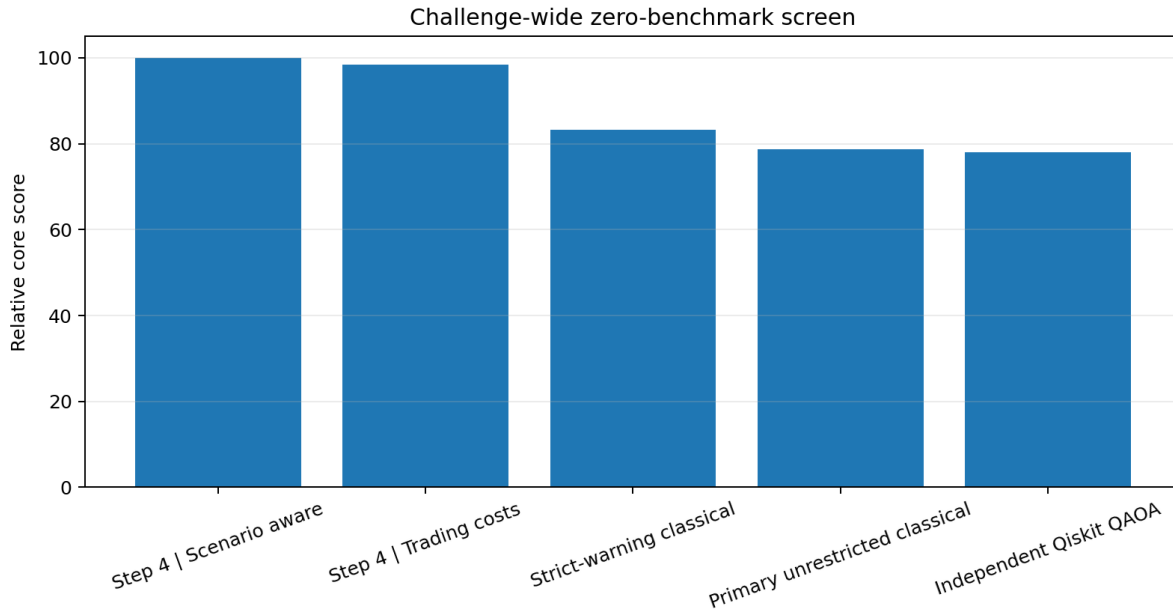
The core score is

$$S_i^{\text{core}} = 100 \cdot \frac{\tilde{A}_i^\sigma + \tilde{A}_i^S + \tilde{A}_i^D}{3}. \quad (8)$$

It is a relative candidate-set score. A value near 100 does not mean a 100% probability of success or an official external grade.

## 6 Challenge-wide zero-benchmark result

| Rank | Portfolio                      | Independent validation | Net return % | Return/vol | Return/stress | Return/drawdown | Relative score |
|------|--------------------------------|------------------------|--------------|------------|---------------|-----------------|----------------|
| 1    | Step 4   Scenario aware        | PENDING_STEP7          | 5.3585       | 0.9028     | 0.5171        | 1.0226          | 99.99          |
| 2    | Step 4   Trading costs         | PENDING_STEP7          | 5.4192       | 0.9027     | 0.5172        | 0.9747          | 98.43          |
| 3    | Strict-warning classical       | PASS                   | 5.4787       | 0.7299     | 0.4502        | 0.8389          | 83.31          |
| 4    | Primary unrestricted classical | PASS                   | 5.4876       | 0.6980     | 0.4317        | 0.7716          | 78.75          |
| 5    | Independent Qiskit QAOA        | PASS                   | 5.4485       | 0.6923     | 0.4285        | 0.7642          | 78.08          |



**Figure 1:** Relative core scores among unique zero-hard-breach candidates.

The Step 4 scenario-aware portfolio is the provisional zero-benchmark leader. It is almost tied with the trading-cost portfolio on return-to-volatility and return-to-stress, but has a materially better return-to-drawdown ratio. Its advantage therefore comes mainly from modeled drawdown reduction.

## 7 Weight sensitivity

Step 10 evaluates 66 combinations

$$(w_\sigma, w_S, w_D) \in \{0, 0.1, \dots, 1\}^3, \quad w_\sigma + w_S + w_D = 1. \quad (9)$$

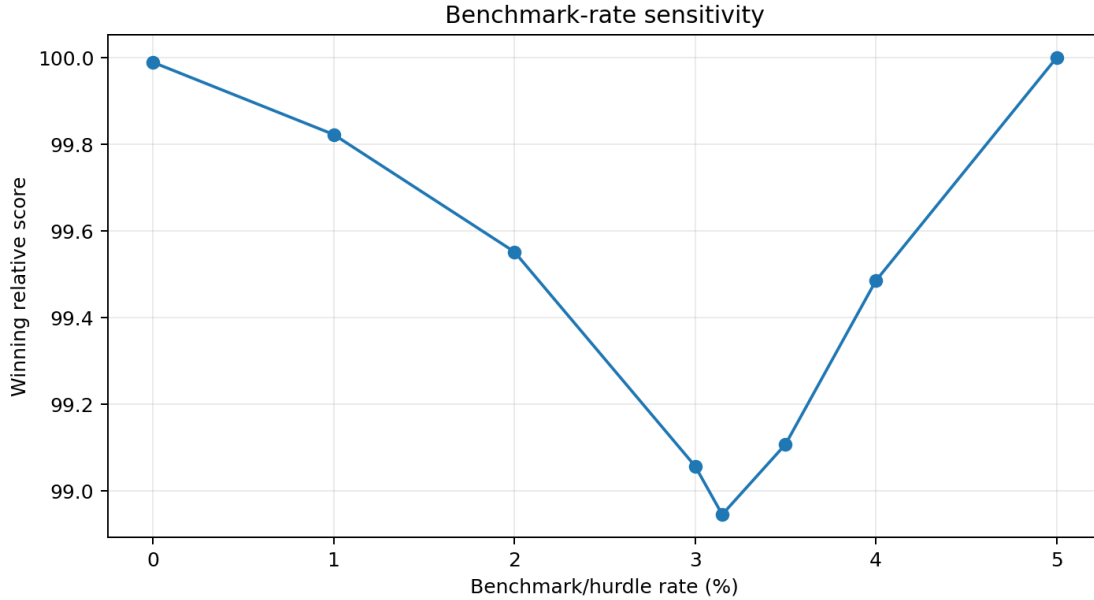
Scenario aware wins 58 of 66 combinations; trading costs wins 8 combinations, primarily when drawdown receives no weight and stress-adjusted return dominates. Within the certified shortlist, strict warning wins all 66 combinations.

## 8 Benchmark-adjusted sensitivity

If a benchmark or hurdle  $r_b$  is declared, the ratios become

$$\frac{R_i^{\text{net}} - r_b}{\sigma_i}, \quad \frac{R_i^{\text{net}} - r_b}{L_i^{\text{worst}}}, \quad \frac{R_i^{\text{net}} - r_b}{\text{MDD}_i}. \quad (10)$$

The project reports a grid of benchmark rates. The provisional leader changes from scenario aware to trading costs above approximately 3.15% in the declared composite sensitivity. This is why the benchmark must be chosen before viewing the winner.



**Figure 2:** The winning score changes as the hurdle rate changes.

## 9 Fully certified shortlist

The three Step 7 certified candidates also have common 1,000-path forward summaries. Their supplemental evidence uses fifth-percentile return, 95th-percentile drawdown, and loss-path frequency.

| Rank | Portfolio                      | Core score | 5th return % | 95th drawdown % | Loss frequency % | Forward downside score |
|------|--------------------------------|------------|--------------|-----------------|------------------|------------------------|
| 1    | Strict-warning classical       | 100.00     | -2.66        | 13.96           | 13.50            | 100.00                 |
| 2    | Primary unrestricted classical | 94.50      | -3.15        | 14.91           | 14.10            | 65.64                  |
| 3    | Independent Qiskit QAOA        | 93.70      | -3.19        | 14.93           | 14.60            | 61.99                  |

Strict warning is the fully certified shortlist leader. It gives up very little expected return while improving all reported downside measures and eliminating soft-warning breaches.

## 10 Why two leaders are reported

The challenge-wide screen is fair to every unique zero-breach project portfolio, but the Step 4 finalists have not yet received the same independent CVXPY/KKT reconstruction and common forward-path evidence as the Step 5/5Q shortlist. Therefore:

- provisional challenge-wide leader: Step 4 scenario aware;
- fully certified shortlist leader: strict-warning classical;
- universal certified winner: pending equivalent Step 4 validation and forward evidence.

This is not indecision; it is evidence discipline.

## 11 Worked score example

If candidate A has ratios (0.90, 0.52, 1.02) and the maxima across candidates are (0.90, 0.52, 1.02), A receives 100 in all three normalized components. If candidate B has (0.73, 0.45, 0.84), its score is

$$100 \cdot \frac{0.73/0.90 + 0.45/0.52 + 0.84/1.02}{3} \approx 83.3. \quad (11)$$

The calculation is transparent, but the magnitude depends on the candidate set.

## 12 Reproduction checklist

Freeze the candidate universe, verify zero hard breaches from exact audits, remove duplicates, declare the horizon and benchmark, calculate net returns and ratios at full precision, store normalization maxima, run the weight grid, report benchmark sensitivity, and distinguish provisional from fully certified evidence.

## 13 What Step 10 establishes

Step 10 provides a transparent interpretation of the challenge rule. It does not supply an official external formula that the challenge organizer did not provide. Its ranking is conditional on the declared risk measures, equal core weights, zero-benchmark base case, and synthetic-model inputs.

## A Formalizing the challenge rule

The challenge statement has a lexicographic structure. Feasibility is evaluated first, and risk-adjusted outcome second. Let  $H_i$  be hard-breach count. The zero-breach screen is

$$\mathcal{E}_0 = \{i : H_i = 0, i \text{ is unique, and } i \text{ is a declared challenge candidate}\}. \quad (12)$$

A hard-breaching portfolio receives no compensating score. This prevents an infeasible high-return portfolio from winning through arithmetic averaging.

A separate certified set is

$$\mathcal{E}_C = \{i \in \mathcal{E}_0 : i \text{ has independent validation and common forward evidence}\}. \quad (13)$$

The distinction is necessary because the Step 4 trading-cost and scenario-aware portfolios have zero reported hard breaches but have not received the same Step 7 and forward-path treatment as the final shortlist.

## B Net return and horizon

If  $R_i$  is annual expected return,  $C_i$  is a one-time initial trading cost, and  $H$  is a holding horizon in years, a simple annualized adjustment is

$$R_{i,H}^{\text{net}} = R_i - \frac{C_i}{H}. \quad (14)$$

The zero-benchmark screen uses  $H = 1$ . This is a convention, not a forecast. Rebalancing frequency and recurring future costs would require a richer model.

## C Core risk-adjusted ratios

Without a declared risk-free rate, the project uses descriptive efficiency ratios rather than naming them Sharpe or Calmar ratios:

$$A_i^\sigma = \frac{R_i^{\text{net}}}{\sigma_i}, \quad (15)$$

$$A_i^S = \frac{R_i^{\text{net}}}{L_i^{\text{worst}}}, \quad (16)$$

$$A_i^D = \frac{R_i^{\text{net}}}{\text{MDD}_i}. \quad (17)$$

These ratios ask how much modeled expected return accompanies one unit of volatility, worst stress loss, or modeled drawdown.

## D Candidate-set normalization

For higher-is-better ratio  $A$ , normalization is

$$\tilde{A}_i = \frac{A_i}{\max_{j \in \mathcal{E}_0} A_j}. \quad (18)$$

The zero-benchmark core score is

$$S_i = 100 \left( \frac{1}{3} \tilde{A}_i^\sigma + \frac{1}{3} \tilde{A}_i^S + \frac{1}{3} \tilde{A}_i^D \right). \quad (19)$$

The number is relative to the frozen candidate set. A value near 100 does not mean near-perfect investment quality or a probability of success.



## E Worked score calculation

For Step 4 scenario-aware, expected return is 5.3780%, one-time cost is 0.0195%, volatility is 5.9354%, worst stress is 10.3633%, and drawdown is 5.24%. Thus

$$R^{\text{net}} = 5.3780 - 0.0195 = 5.3585\%, \quad (20)$$

$$A^\sigma = 5.3585/5.9354 \approx 0.902804, \quad (21)$$

$$A^S = 5.3585/10.3633 \approx 0.517065, \quad (22)$$

$$A^D = 5.3585/5.24 \approx 1.022615. \quad (23)$$

After normalization, its score is approximately 99.9895. The trading-cost portfolio is almost tied on volatility and stress efficiency but has a lower drawdown-efficiency ratio, so scenario-aware leads under equal weights.

## F Weight sensitivity

Let weights  $(a, b, c)$  satisfy  $a, b, c \geq 0$  and  $a + b + c = 1$ . The generalized core score is

$$S_i(a, b, c) = 100 \left( a\tilde{A}_i^\sigma + b\tilde{A}_i^S + c\tilde{A}_i^D \right). \quad (24)$$

Across 66 declared weight combinations, scenario-aware wins 58 and trading costs wins 8. Trading costs wins cases in which drawdown receives zero weight and stress efficiency is emphasized. This reveals the economic reason for the ranking rather than merely asserting robustness.

## G Benchmark-rate sensitivity

A positive benchmark  $r_b$  changes the numerator:

$$A_i^\sigma(r_b) = \frac{R_i^{\text{net}} - r_b}{\sigma_i}, \quad (25)$$

with analogous stress and drawdown ratios. Because trading costs has slightly higher expected return than scenario-aware, it becomes provisional leader above a benchmark of roughly 3.15% in the declared sensitivity analysis. The benchmark must be predeclared from a cash return or investment hurdle, not chosen after observing the winner.

## H Certified shortlist score

The certified set contains primary classical, strict warning, and QAOA. These candidates share independent validation and common forward-path summaries. Strict warning has nearly the same expected return as primary but lower volatility, stress, drawdown, tail drawdown, loss frequency, and zero soft warnings. It remains first across all 66 certified-set core-weight combinations.

## I Why no universal certified winner is declared

The challenge-wide provisional leader lacks equivalent Step 7 and forward evidence. The correct status is therefore:

Step 4 scenario-aware is the provisional zero-benchmark leader; strict warning is the independently certified shortlist leader; universal certification is pending equivalent validation and common forward evidence for the Step 4 zero-breach candidates.

This avoids both unfairly excluding valid zero-breach portfolios and overstating evidence that has not been produced.

## J Necessary next validation

To certify a universal winner, apply the Step 7 independent QP reconstruction, KKT checks, full constraint audit, objective reconciliation, and common 1,000-path simulation to Step 4 trading costs and scenario-aware. Use full-precision inputs rather than rounded display values. Then freeze the benchmark and scoring weights before recalculating the leaderboard.

## K Automated integrity tests

- no positive-hard-breach row can receive a score;
- baselines and duplicates cannot receive challenge votes;
- net return reconciles to declared horizon and cost;
- percentages and decimal fractions cannot mix;
- score weights sum to one;
- metric directions are correct;
- winner equals the maximum score rather than a hardcoded name;
- the status certificate matches evidence availability; and
- narratives do not call synthetic results historical or claim quantum advantage.

## L Reproduction protocol

1. Import full-precision candidate metrics and eligibility metadata.
2. Apply the zero-hard-breach and duplicate gates.
3. Calculate net return under declared horizon and benchmark.
4. Compute the three core ratios and candidate-set normalization.
5. Run weight and benchmark sensitivities.
6. Separately score the fully certified set with common forward evidence.
7. Export scorecards, eligibility audit, winner frequencies, and a status certificate.
8. Refuse to label the provisional winner universally certified until evidence parity is achieved.

## Scoring and provenance clarification

These notes align interpretation with the final executed code path and saved evidence without changing the reported numerical results.

- The score is a bespoke, transparent candidate-set normalization. It is not an externally supplied official formula or an established institutional performance statistic.
- The standalone scorer uses a frozen candidate-input table assembled from upstream saved results. It reproduces the scoring arithmetic, but the current release does not claim fully automated upstream-to-Step-10 lineage.
- The Step 4 provisional leaders use the full 50% turnover allowance and do not yet share the same independent Step 7/common-forward evidence as the certified shortlist.
- A universal winner therefore remains conditional on equal validation depth, a predeclared benchmark, and further mandate sensitivities.
- Exact reduced-QUBO support does not by itself prove exhaustive exact-finance support optimality across all 462 continuously refined supports.